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Utilizing Smartphone-derived Photogrammetry 3D Model for AI-based Riparian Crack Segmentation and Measurement

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Abstract. In this paper, the authors present a simple free methodology that harnesses the potential of smartphone-derived photogrammetry 3D models for the AI-driven segmentation and quantification of riparian cracks. The approach the authors propose leverages the YOLOv7-seg model to precisely delineate cracks, employing photogrammetry techniques to construct an orthophoto, subsequently enabling the extraction and accurate measurement of crack dimensions. This research underscores the fact that the proposed methodology yields outcomes in both the detection and measurement aspects of riparian cracks. This investigation carries profound implications for the realm of environmental oversight and administration, as it heralds an innovative approach toward the identification and amelioration of issues pertaining to cracks in riparian asphalt surfaces. Through the rigorous analysis of this approach, not only crack detection and measurement but also holds promise for wider applications in the field of environmental assessment. It extends the boundaries of AI-driven methodologies in the context of environmental preservation.

Keywords: Instance Segmentation, Riparian Crack Detection, River Monitoring, Smartphone

1 Introduction

The monitoring of cracks along river banks is a vital aspect of the evaluation of the environmental stabilization of riparian areas. Recently, promising progress has been made in riverbank crack monitoring by using Unmanned Aerial Vehicle (UAV) platforms for artificial intelligence tasks such as instance segmentation [1]. However, high accurate instance segmentation using closer distance has not been thoroughly verified and evaluated. To address this gap, the authors collected several angle-fixed videos along the targeted river bank for selecting the suitable one, and this video was used to be inferred by the You Only Look Once version 7 (YOLOv7) instance segmentation model specifically tailored for crack segmentation, referred to as YOLOv7-seg. In this research, the authors generated a work flow that the asphalt-paved cracks can be extracted and size-measured automatically, and by a free version 3D model processing software, the video-derived frames can be transformed to an orthophoto, which these results can be important data accumulation for the construction ecosystem digital-twin in the future.

2 Work flow

A systematic process for measuring crack size using various technologies and methods is outlined in the workflow shown in the Figure 1. It starts with the acquisition of videos using a smart phone with a fixed camera angle of 45°. These video are then processed by Video2Eachframe.py to be analyzed by YOLOv7-seg Inference. Simultaneously, the images are used to create a 3D texture mesh, which is further refined by photogrammetry using 3DF Zephyr. Orthophoto is used along with measurement programming to facilitate accurate crack size measurements. The final result provides detailed cracked measurements.

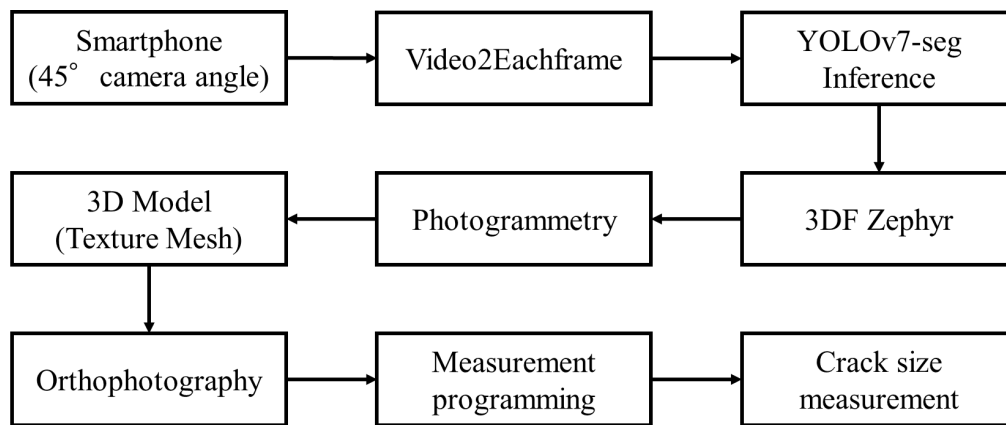


Figure 80 Process of the data collection and the workflow .

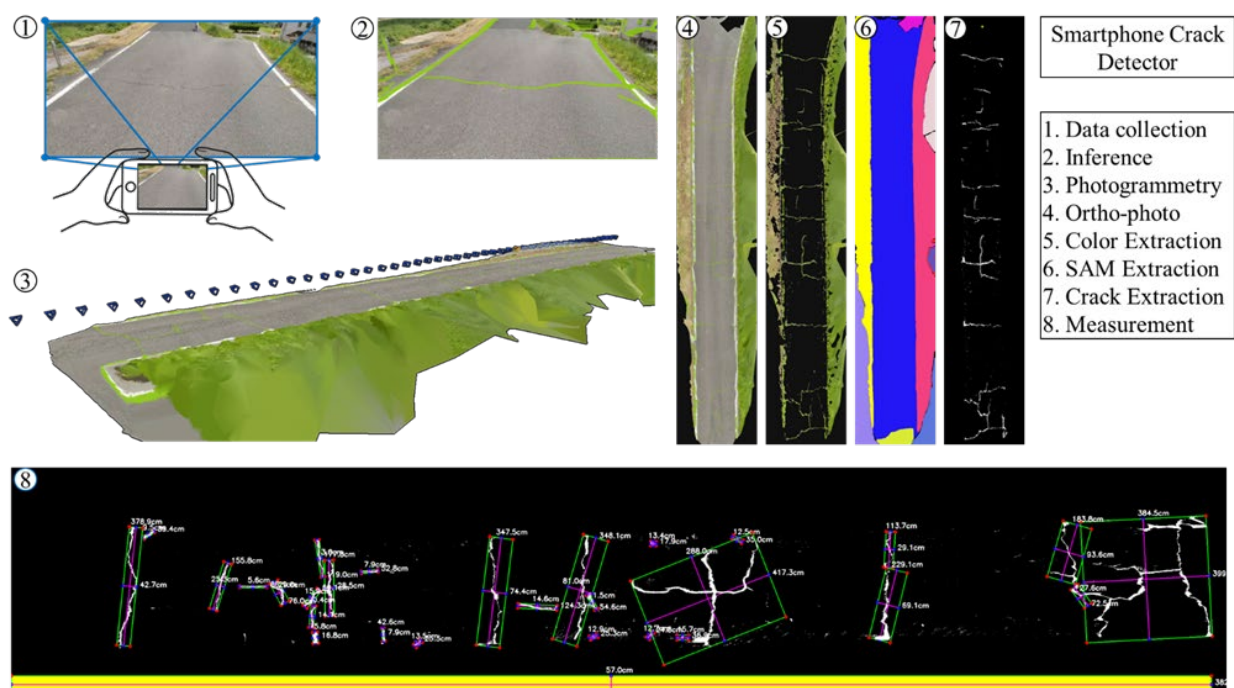


Figure 2 Process of the crack size measurement

3 Conclusion

As shown in Figure 2, this study has extracted the cracks with the detailed information from the video after the mentioned process. In this process, orthophoto ensures accurate surface representation, while measurement programming facilitates precise crack size determination. This comprehensive approach yields detailed measurements, enhancing our understanding of cracks and aiding in maintenance and safety assessments.

Reference

- [1] Shijun, P. A. N., YOSHIDA, K., & NISHIYAMA, S. Detection and Segmentation of Riparian Asphalt Paved Cracks Using Drone and Computer Vision Algorithms. *Intelligence, Informatics and Infrastructure*, 4(2), 35-49. 2023